1 **(a)** Translation **[B1]**

by vector $\begin{pmatrix}4\\0\end{pmatrix}$ **[B1]**

(b) $f(x)+7$ **[B1]**

[Total Marks: 3]

2 **(a)** x^2-5 **[B1]**

(b) $(x-3)^2$ **[B1]**

(c) Translation **[B1]**

by vector $\begin{pmatrix}-4\\-1\end{pmatrix}$ **[B1]**

[Total Marks: 4]

3 **(a)** $g(x)=f(x)+7$ **[M1]**

$g(x)=2x+2$ **[A1]**

(b) Translation **[B1]**

by vector $\begin{pmatrix}4\\0\end{pmatrix}$ **[B1]**

[Total Marks: 4]

4 **(a)** Horizontal shift 3 seen **[B1]**

$\begin{pmatrix}3\\-2\end{pmatrix}$ **[B1]**

(b) $(x-3)^2$ seen **[B1]**

$(x-3)^2-2$ **[B1]**

(c) $x=-5$ **[B1]**

$(-5,-1)$ **[B1]**

[Total Marks: 6]

5 **(a)** $(-2,1)$ **[B1]**

(b) $(5,-1)$ **[B1]**

(c) **(i)** $a=2$ **[B1]**

$b=-5$ **[B1]**

(ii) $(5,-6)$ **[B1]**

(Strict follow through from their vector in (c)(i) applied to $(3,-1)$)

[Total Marks: 5]

6 **(a)** $g(x) = f(x+2) = 3(x+2)^2$ **[M1]**

$3(x^2 + 4x + 4)$ **[M1]**

$3x^2 + 12x + 12$ **[A1]**

(b) $(-2, -8)$ **[B2]**

(Allow B1 for each correct coordinate)

(c) Substitutes $x = 0$ into $h(x) = 3(x+2)^2 - 8$ (or similar valid method) **[M1]**

4 **[A1]**

[Total Marks: 6]

7 **(a)** Translation by 2 units left or 3 units down **[M1]**

Correct line segments joining $(-3, -2)$, $(-1, 1)$, $(1, -2)$, and $(2, -1)$ **[A1]**

(b) Translation **[B1]**

Vector $\begin{pmatrix} 0 \\ -5 \end{pmatrix}$ **[B1]**

(c) $f(x) - 5$ **[B1]**

(d) $x = 4 + 3 = 7$ or $y = 2 + 4 = 6$ **[M1]**

$(7, 6)$ **[A1]**

[Total Marks: 7]

8 **(a)** $(4, -1)$ **[B1]**

(b) $f(x+3) + 2$ **[B2]**

(B1 for $f(x+3)$ and B1 for $+2$)

(c) $f(-k) = 0$, so $-k = 2$ or $-k = 6$ **[M1]**

$k = -2$ and $k = -6$ **[A1]**

(d) $f(x-3)$ **[B1]**

[Total Marks: 6]

9 **(a)** $g(x) = f(x+4)$ **[M1]**

$(x+4)^2 - 2(x+4) - 3$ **[M1]**

$x^2 + 8x + 16 - 2x - 8 - 3$ **[M1]**

$x^2 + 6x + 5$ **[A1]**

(b) $h(x) = x^2 - 2x - 3 - 5 = x^2 - 2x - 8$ **[M1]**

$(x-4)(x+2) = 0$ **[M1]**

$x = 4$ and $x = -2$ **[A1]**

[Total Marks: 7]

10 **(a)** Finds $(x-3)^2$ as part of completing the square: $f(x) = (x-3)^2 - 9 + 14$ **[M1]**

$f(x) = (x-3)^2 + 5$ **[M1]**

Translation vector $\begin{pmatrix} 3 \\ 5 \end{pmatrix}$ **[A1]**

(b) Substitutes $x-4$ into $g(x)$: $g(x-4) = \dfrac{3}{(x-4)+2} - 5$ **[M1]**

Adds 2 to the function: $h(x) = \dfrac{3}{x-2} - 5 + 2$ **[M1]**

$h(x) = \dfrac{3}{x-2} - 3$ or equivalent single fraction $\dfrac{9-3x}{x-2}$ **[A1]**

(c) $x = 2$ **[B1]**

$y = -3$ **[B1]**

(Strict follow through from their $h(x)$ in part (b))

[Total Marks: 8]

11 **(a)** **(i)** $\dfrac{20}{x} - 4$ **[B1]**

(ii) $\dfrac{20}{x} - 4 = 0$ **[M1]**

$x = 5$ **[A1]**

(b) $h(x) = f(x-k) = \dfrac{20}{x-k}$ **[M1]**

$\dfrac{20}{8-k} = 5$ **[M1]**

$k = 4$ **[A1]**

(c) Vertical asymptote: $x = 3$ **[B1]**

Horizontal asymptote: $y = 7$ **[B1]**

[Total Marks: 8]

12 **(a)** **(i)** $f(x+2) - 6$ **[B1]**

(ii) $2(x+2)^2 - 5(x+2) + 1 - 6$ **[M1]**

$2(x^2 + 4x + 4) - 5x - 15$ **[M1]**

$2x^2 + 3x - 7$ **[A1]**

(b) $2(x-p)^2 - 5(x-p) + 1 + q = 2x^2 - 13x + 18$ **[M1]**

$-4p - 5 = -13$ giving $p = 2$ **[A1]**

$2(2)^2 + 5(2) + 1 + q = 18$ giving $q = -1$ **[M1]**

$\begin{pmatrix} 2 \\ -1 \end{pmatrix}$ **[A1]**

(c) $2x^2 - 5x + 1 = 2x^2 + 3x - 7$ **[M1]**

$8x = 8$ giving $x = 1$ **[A1]**

$(1, -2)$ **[A1]**

[Total Marks: 11]